PEACOCK
DYNAMICS

MARINE AUTONOMY - SWARM CONTROL - ROBOTIC EXPLORATION

SESSION 01

FOUNDATIONS OF AUTONOMOUS SYSTEMS

Dynamical systems, state-space modeling, controllability, and observability.

ACTIVE TRACK
ALGORITHM-FIRST

MISSION CONTEXT

Robots are systems, not screenshots

Every robot evolves over time according to physics.
Control theory describes, predicts, influences, and stabilizes that evolution.
Before swarms, autonomy, or AI, we must understand one vehicle.

ROBOT → PHYSICS → CONTROL

One vehicle first.
Many vehicles later.

CORE IDEA

A dynamical system changes state over time

CURRENT STATE
$x(t)$

→

DYNAMICS
RULES OF MOTION

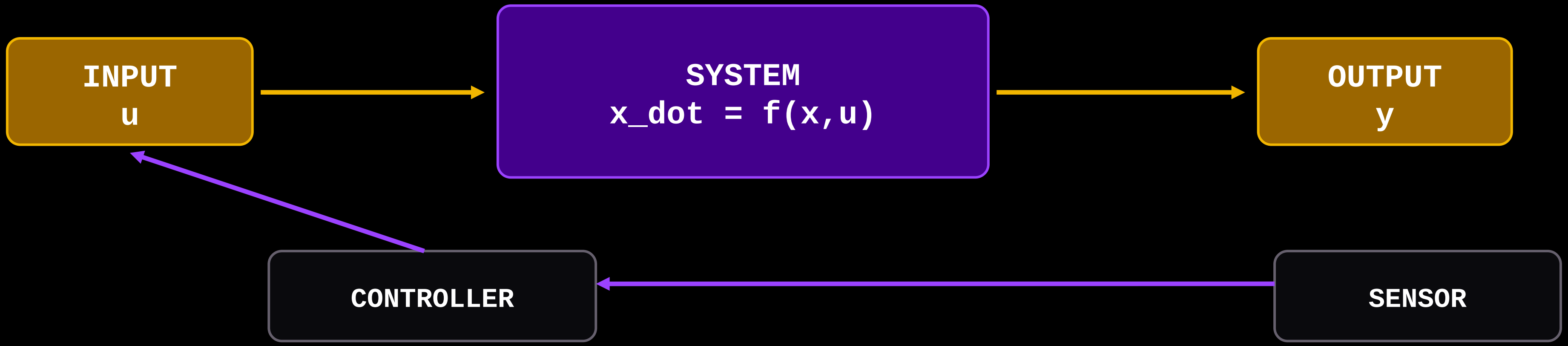
→

NEXT STATE
$x(t + dt)$

State = enough information to predict what happens next.

INPUT - STATE - OUTPUT

Control begins with information flow



Example:
u = propeller thrust
x = position + velocity + heading
y = GPS / compass / camera reading

MODEL CLASSES

Real robots are nonlinear. We approximate locally.

LINEAR MODEL

x_dot = A x + B u

Superposition works
Easier to analyze and design

NONLINEAR MODEL

x_dot = f(x, u)

Drag, saturation, friction, geometry
Harder, but more realistic

Linear control is not denial of nonlinearity. It is a local engineering weapon.

WHY LINEARIZATION?

Make a hard system locally useful

OPERATING POINT
(x0, u0)

→

SMALL DEVIATION
dx, du

→

LINEAR MODEL
dx_dot = A dx + B du

```
For nonlinear dynamics:
x_dot = f(x,u)

Around (x0,u0):
delta_x_dot = A delta_x + B delta_u

A = df/dx |(x0,u0)
B = df/du |(x0,u0)
```

Pendulum linearization

From sin(theta) to theta

Nonlinear equation:

theta_ddot + (g/L) sin(theta) = tau / (m L^2)

This is nonlinear because sin(theta) is nonlinear.

Small-angle assumption:

theta approx 0
sin(theta) approx theta

Valid when theta is small, e.g. < 10-15 degrees.

Physical meaning: near the hanging equilibrium, the pendulum behaves almost like a linear oscillator.

Pendulum: local linear model

Same system, simpler mathematics

Step 1: Start nonlinear
theta_ddot + (g/L) sin(theta) = tau/(m L^2)

Step 2: Small angle
sin(theta) approx theta

Step 3: Linearized equation
theta_ddot + (g/L) theta = tau/(m L^2)

State choice:
x1 = theta
x2 = theta_dot

Then:
x1_dot = x2
x2_dot = -(g/L)x1 + (1/(mL^2)) tau

Takeaway: linearization turns nonlinear physics into state-space form.

Modern control language

Put dynamics into matrix form

```
Continuous-time LTI system:

x_dot = A x + B u
y     = C x + D u
```

x: internal state vector
u: control input vector
y: measured output vector
A, B, C, D: matrices defining dynamics and measurement

This form powers LQR, MPC, observers, Kalman filters, and swarm control.

What do A, B, C, D mean?

The four matrices are not decorative

A MATRIX
Natural system dynamics
How states affect each other

Example:
velocity changes position

B MATRIX
Input influence
How actuators push states

Example:
thruster changes velocity

C MATRIX
Sensor map
Which states are measured

Example:
GPS sees position

D MATRIX
Direct feedthrough
Input directly affects output

Often zero in robotics

Simple 1D marine robot

Position and velocity as state

```
State:
x = [ position ; velocity ]

Input:
u = thrust

Output:
y = position measurement
```

```
x_dot = A x + B u

A = [ 0    1 ]
    [ 0  -b/m ]

B = [ 0 ]
    [ 1/m ]

C = [ 1  0 ]   D = [ 0 ]
```

This is intentionally simple. The point is structure, not hydrodynamic perfection.

Can we move it?

Influence is the first control question

INITIAL STATE $x0$ → **CONTROL INPUTS** $u(t)$ → **TARGET STATE** $x1$

A system is controllable if suitable inputs can drive it from one state to another over time.

Matrix test, intuition first

Does input reach every state direction?

For n states:

Cmat = [B   A B   A^2 B   ...   A^(n-1) B]

Controllable if:
rank(Cmat) = n

Interpretation:

B tells where input enters.
AB tells where input goes after dynamics.
A^2B continues propagation.

Full rank means input reaches all state directions.

Do not worship the rank test. Use it as a diagnostic: can the actuator influence the whole state?

Can we know it?

Information is the first estimation question

TRUE STATE
x

→

SENSORS
y = Cx + Du

→

ESTIMATED STATE
x_hat

Observable means the hidden internal state can be reconstructed from measured outputs over time.

Matrix test, intuition first

Do measurements reveal every state direction?

```
For n states:

Omat = [   C
           C A
           C A^2
           ...
           C A^(n-1) ]

Observable if rank(Omat) = n
```

```
Interpretation:

C tells what the sensor directly sees.
CA tells what dynamics reveal next.
CA^2 keeps unfolding hidden state information.

Full rank means all states can be inferred.
```

Robotics translation: if you cannot measure it directly, can the motion history reveal it?

Two different questions

Influence and knowledge are not the same

CONTROLLABILITY	OBSERVABILITY
Question: Can inputs move the state?	Question: Can outputs reveal the state?
Matrix pair: (A, B)	Matrix pair: (A, C)
Failure: Actuator cannot affect some mode	Failure: Sensor cannot see some mode

Basic controller form

Use state to compute action

STATE x → **GAIN MATRIX** K → **CONTROL** $u = -Kx$ → **PLANT** $\dot{x} = Ax + Bu$

```
Closed-loop dynamics:

x_dot = (A - B K) x
```

What does K do?

It reshapes the closed-loop behavior

```
Open-loop:
x_dot = A x + B u

Feedback:
u = -Kx

Closed-loop:
x_dot = (A - BK)x
```

Stabilize unstable motion
Reduce tracking error
Reject disturbances
Trade speed against effort
Prepare for LQR and MPC later

For today: K is not magic. It is a gain matrix that changes system behavior.

Single-agent control becomes swarm autonomy

Coordination layers sit above local control

ONE VEHICLE
state x_i

→

LOCAL CONTROL
u_i

→

COMMUNICATION
neighbor states

→

SWARM POLICY
coverage + spacing

Consensus intuition:

u_i depends on x_i and neighboring x_j

Local stability + coordination = useful swarm behavior

Where this fits in the company

Control theory is the spine of autonomy

Local robot control uses controllability.
State estimation uses observability.
Trajectory tracking uses feedback.
Formation control uses multi-agent dynamics.
Consensus uses state exchange between robots.

TECHNICAL LINEAGE
Simulation -> Control -> Estimation -> Coordination -> Marine Swarms

Minimum vocabulary to remember

Enough to join future discussions

x: state
u: input
y: output
A: natural dynamics
B: actuator influence
C: sensor mapping
D: direct input-output path
K: feedback gain

Controllability: can we move it?
Observability: can we know it?
Linearization: local approximation
State feedback: u = -Kx
Closed loop: x_dot = (A-BK)x

What should survive after the lecture?

The core mental model

Every robot is a dynamical system.
State-space is the language of modern control.
Controllability asks whether inputs can move the system.
Observability asks whether sensors reveal the state.
Swarm autonomy is built on single-agent control.

If they remember only two lines:
Controllability = can we move it?
Observability = can we know it?

PEACOCK DYNAMICS

FROM SINGLE-AGENT CONTROL TO AUTONOMOUS MARINE SWARMS

The control foundation is now loaded. Next: LQR, MPC, estimation, and swarm coordination.

संप्रभुता । एकत्वम् । नियन्त्रणम् ।